SpaceX Falcon 9 First-Stage Landing Prediction

Data Science Capstone Project

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IBM Data Science Professional Certificate – Coursera

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GitHub Repository: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone

Executive Summary

Objective

- Predict whether the Falcon 9 first stage will land successfully using historical launch, payload, orbit, launch-site, and booster characteristics.

Data and Methodology

- The project combines SpaceX API data, a saved Wikipedia launch table snapshot, data wrangling, Python visualization, ten SQL queries, Folium geospatial analysis, a Plotly Dash application, and classification modeling.

Exploratory Findings

- Landing success is more frequent among later flights and varies across launch sites and orbit types. Payload mass alone does not completely separate successful and unsuccessful outcomes.

Predictive Modeling

- Logistic Regression, SVM, Decision Tree, and KNN were tuned using five-fold GridSearchCV. Decision Tree achieved the highest mean cross validation accuracy of 86.3% and a test accuracy of 77.8%.

Business Value

- **Landing-success prediction can support preliminary mission-risk assessment and launch-cost estimation by indicating the likelihood of first-stage recovery.**

Introduction

- **Project Background:** SpaceX advertises Falcon 9 launches at approximately USD 62 million, which is substantially lower than many traditional launch alternatives. A major source of cost savings is the ability to recover and reuse the first stage.
- **Business Problem:** The cost of a launch depends partly on whether the first stage can be recovered. Predicting the landing outcome can therefore support launch-cost and mission-risk estimation.
- **Data Science Question:** Can launch characteristics such as payload mass, launch site, orbit type, booster information, and flight history be used to predict whether the Falcon 9 first stage will land successfully?
- **Target Variable**
  Class = 1: successful first-stage landing. Class = 0: unsuccessful or non-successful landing outcome
- **Completed Project Deliverables:**
  - ✓ SpaceX API data collection — Slide 4
  - ✓ Wikipedia web scraping — Slide 5
  - ✓ Data wrangling — Slide 6
  - ✓ EDA visualizations — Slides 7–9
  - ✓ SQL analysis — Slides 10–16
  - ✓ Folium analysis — Slides 17–19
  - ✓ Plotly Dash — Slides 20–21
  - ✓ Predictive Analysis — Slides 22–24
  - ✓ Conclusion — Slide 25
  - ✓ GitHub Repository Contents — Slide 26

Data Collection – SpaceX API

- **Data Source**
  Launch information was retrieved from the SpaceX REST API using requests.get().
- **Processing Steps**
  1. Retrieved launch records in JSON format.
  2. Used pandas.json_normalize() to flatten nested JSON objects.
  3. Extracted relevant variables, including flight number, launch date, payload mass, orbit, launch site, booster information, and landing outcome.
  4. Filtered the records to retain Falcon 9 launches only.
  5. Replaced missing payload-mass values with the mean of the observed payload masses.

GitHub URL for API Notebook:
https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection – Web Scraping

Data Source

The notebook defines a function for reading Falcon 9 and Falcon Heavy launch tables from Wikipedia using pandas.read_html().

For the saved execution, live web scraping was disabled and the IBM Skills Network course snapshot was used.

Processing Steps

1. Read and flattened the launch table headers.
2. Detected tables containing flight and launch information.
3. Used the IBM course snapshot when live scraping was disabled or unavailable.
4. Standardized column names and whitespace.
5. Saved 101 launch records to spacex_web_scraped.csv.

GitHub URL for Web Scraping Notebook: *https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb*

Data Wrangling and Feature Preparation

Verified Results

- Input dataset: 90 Falcon 9 launch records.
- Missing values: 26 missing LandingPad values.
- Class = 1 was assigned to successful landing outcomes.
- Class = 0 was assigned to unsuccessful or non-successful outcomes.
- Successful outcomes: 60 launches (66.7%).
- Unsuccessful outcomes: 30 launches (33.3%).
- The labeled dataset was saved as *https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/dataset_part_2.csv*.

GitHub URL for Data Wrangling Notebook:
https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb

EDA with Data Visualization

Exploratory Questions

- Does landing success become more frequent as the flight number increases?
- How does payload mass relate to landing outcome?
- Do landing-success rates differ by launch site and orbit type?
- How has landing success changed over time?

Visualizations

- Scatter plots: flight number and payload mass versus landing outcome.
- Site-level plots: landing success across launch sites.
- Bar charts: success rate and launch count by orbit type.
- Line charts: annual landing-success trend.

Pandas and Matplotlib

Pandas, Matplotlib.

Feature engineering: 90 rows, 80 predictors, *https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/dataset_part_3.csv*

GitHub URL for EDA Visualization Notebook:

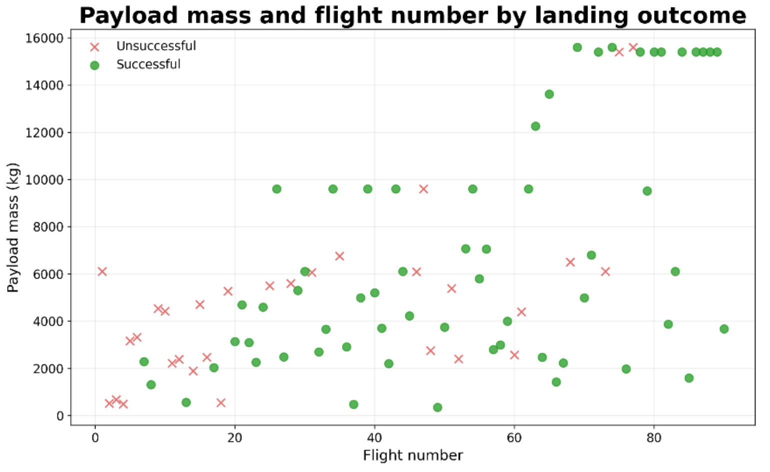
https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/eda_visualization.ipynb

Landing Outcome by Flight Number and Payload Mass

Key Findings:

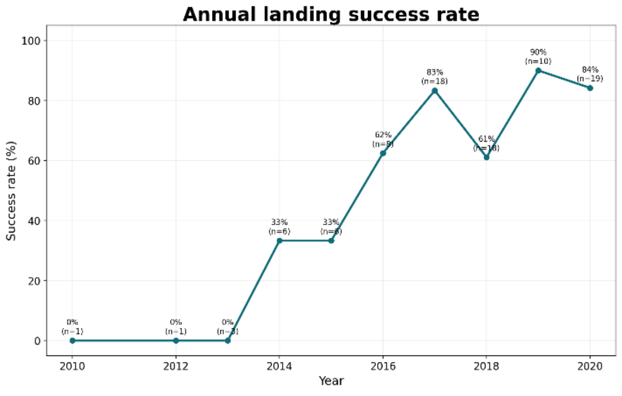
- Successful landings become more frequent among later Falcon 9 flights, which is consistent with operational improvement over time.
- Payload mass alone does not perfectly separate successful and failed landings.
- Landing outcome should therefore be analyzed together with launch site, orbit type, and flight history.

Interpretation Note: These relationships are observational and should not be interpreted as proof of causality.



Later flights contain a higher concentration of successful landings; payload mass alone does not fully separate outcomes.

Annual Landing Success Trend



Success increased markedly after 2016, although annual rates fluctuate because yearly launch counts are limited.

- Landing success was very low during the earliest years in the dataset.
- The annual success rate increased substantially after 2014.
- The rate reached approximately 83% in 2017 and 90% in 2019.
- Annual percentages should be interpreted together with the number of launches in each year.

EDA with SQL

SQL Analysis Environment
The launch dataset was loaded into a SQLite database and analyzed through ten completed SQL tasks in a Jupyter Notebook. The queries used SELECT DISTINCT, LIKE, BETWEEN, aggregate functions, grouping, sorting, date filtering, and subqueries. Query code and outputs are presented on Slides 11–16.

Analysis Objectives

- Identify the unique launch sites.
- Calculate total and average payload masses.
- Analyze mission and landing outcomes.
- Identify the first successful ground-pad landing.
- Compare booster versions within selected payload ranges.
- Analyze landing outcomes across time periods.

SQL Techniques: SELECT DISTINCT, WHERE, LIKE, BETWEEN, GROUP BY, ORDER BY, COUNT, SUM, AVG, MIN, and subqueries.

Completed SQL Notebook: GitHub URL for SQL Notebook: *https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/jupyter_labs_eda_sql_coursera_sqllite.ipynb*

EDA with SQL Results

Task 1

Display the names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

EDA with SQL Results

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

Total_Payload_Mass
45596

EDA with SQL Results

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Average_Payload_Mass
2928.4

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%sql SELECT MIN(Date) AS First_Successful_Landing_Date FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

First_Successful_Landing_Date
2015-12-22

EDA with SQL Results

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) as Total_Number FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total_Number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA with SQL Results

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%sql SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

EDA with SQL Results

Task	Question	Result
Task 1	Unique launch sites	4 sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Task 2	Launch sites beginning with CCA	First five matching records displayed in the SQL notebook.
Task 3	Total payload for NASA (CRS)	45,596 kg
Task 4	Average payload for F9 v1.1	2,928.4 kg
Task 5	First successful ground-pad landing	2015-12-22
Task 6	Successful drone-ship boosters with payload 4,000–6,000 kg	4 records: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
Task 7	Mission outcomes	98 standard successes; 1 in-flight failure; 2 other success labels.
Task 8	Maximum payload	15,600 kg across 12 booster records.
Task 9	Drone-ship failures in 2015	2 records: 2015-01-10, 2015-04-14
Task 10	Landing outcomes, 2010-06-04 to 2017-03-20	No attempt: 10; Failure (drone ship): 5; Success (drone ship): 5; Controlled (ocean): 3; Success (ground pad): 3; Failure (parachute): 2; Uncontrolled (ocean): 2; Precluded (drone ship): 1

Folium Methodology

Folium Geospatial Analysis

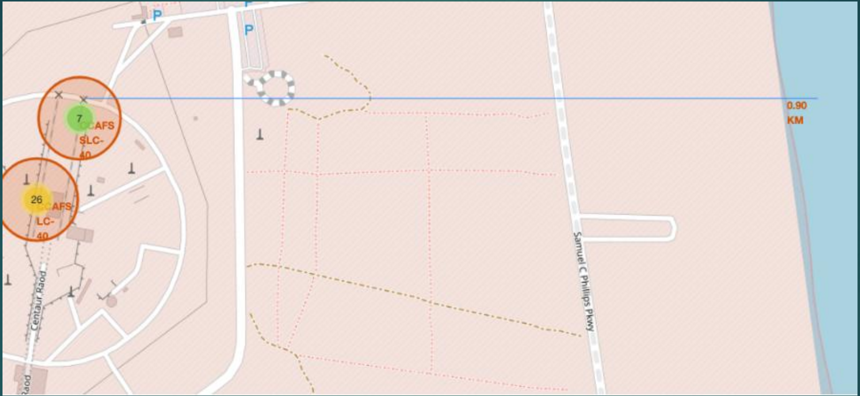
- Plotted all launch-site locations on an interactive map.
- Added labels for each launch site.
- Used color-coded markers to distinguish successful and unsuccessful landings.
- Applied MarkerCluster to organize overlapping launch markers.
- Measured distances from launch sites to coastlines, roads, railways, and nearby cities.

Purpose

Examine whether launch sites share common geographic and infrastructure characteristics.

GitHub URLs: *https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb*

Folium Findings and Site MetricsFolium Map Results



- The four launch sites are located near the coast.
- Three launch-site labels are located in Florida, while VAFB SLC-4E is located in California.
- Coastal locations support safe launch trajectories over the ocean.

Folium Findings and Site Metrics

Launch Site	Latitude	Longitude	Launches	Successful	Success rate
CCAFS LC-40	28.56230197	-80.57735648	26	7	26.9%
CCAFS SLC-40	28.56319718	-80.57682003	7	3	42.9%
KSC LC-39A	28.57325457	-80.64689529	13	10	76.9%
VAFB SLC-4E	34.63283416	-120.6107455	10	4	40.0%

- KSC LC-39A recorded the highest success rate at 76.9% in this dataset.
- CCAFS LC-40 had the largest number of launches but the lowest success rate at 26.9%.
- Launch count and success rate should be considered together when comparing sites.

Plotly Dash Methodology

Plotly Dash Application

- Added a launch-site dropdown selector.
- Added a payload-mass range slider.
- Created a pie chart showing landing outcomes by launch site.
- Created a scatter plot showing payload mass versus landing outcome.
- Used booster-version categories to compare performance across mission configurations.

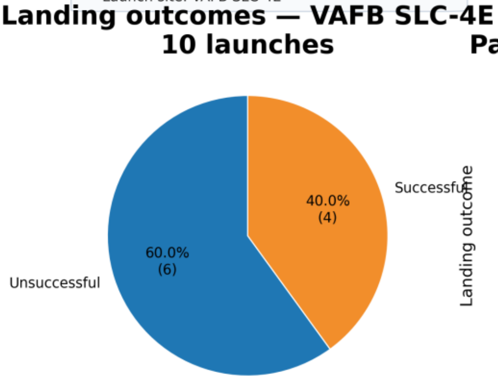
The dashboard was implemented in spacex_dash_app.py using Dash callbacks to update both charts based on the selected launch site and payload range.

GitHub URL for Plotly Dash Application:

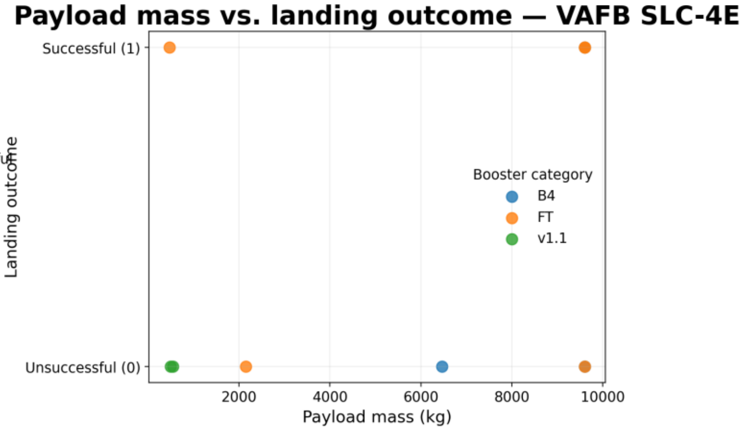
https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_dash_app.py

Plotly Dash Results

Launch site: VAFB SLC-4E



Payload range: 0–9,600 kg



Key result: VAFB SLC-4E has 4 successful and 6 unsuccessful outcomes in the dashboard dataset (40.0% success).

Predictive Analysis — Classification Methodology

Dataset
- 90 launch records.
- 80 engineered predictor columns.
- Target: Class, where 1 = successful and 0 = unsuccessful.

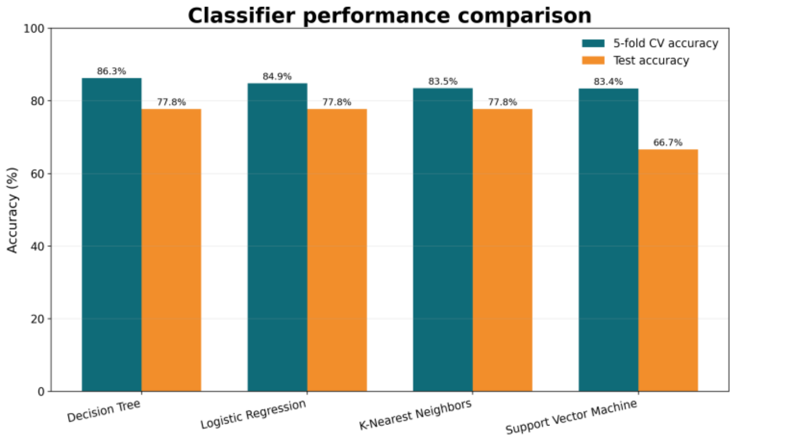
Preprocessing
- Standardized predictors using **StandardScaler.**
- Used a stratified 80/20 train-test split.
- Training set: 72 records.
- Test set: 18 records.
- random_state = 2.

Models
- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Model Tuning
- GridSearchCV with five-fold cross validation.
- Primary tuning metric: accuracy.

Evaluation
- Mean cross validation accuracy
- Test accuracy
- Precision, recall, and F1-score
- Confusion matrix

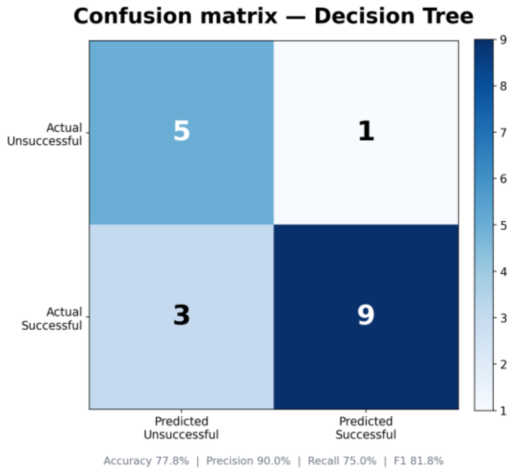
Predictive Analysis Results — Model Comparison



Key Results

- Decision Tree achieved the highest mean five-fold cross validation accuracy: 86.3%.
- Decision Tree, Logistic Regression, and KNN each achieved 77.8% test accuracy.
- SVM achieved a lower test accuracy of 66.7%.
- Decision Tree was selected as the final model based primarily on cross validation performance.

Predictive Analysis Results — Confusion Matrix



Confusion Matrix Interpretation

- 5 unsuccessful landings were correctly classified.
- 9 successful landings were correctly classified.
- 1 unsuccessful landing was incorrectly predicted as successful.
- 3 successful landings were incorrectly predicted as unsuccessful.

Model Metrics

- Accuracy: 77.8%
- Precision: 90.0%
- Recall: 75.0%
- F1-score: 81.8%

Conclusion — Key Findings and Business Implications

1. Exploratory Findings
Landing success becomes more frequent among later Falcon 9 flights. Observed success rates also vary across launch sites and orbit types. Payload mass alone does not fully explain the landing outcome.

2. Data Wrangling
The labeled dataset contains 90 launches: 60 successful and 30 unsuccessful. Feature engineering produced 80 predictors for classification.

3. SQL Findings
The SQL dataset contains four launch-site labels. The average payload mass for F9 v1.1 missions is 2,928.4 kg, and the first successful ground-pad landing occurred on December 22, 2015.

4. Predictive Modeling
Decision Tree achieved the highest mean cross validation accuracy of 86.3% and a test accuracy of 77.8%. Logistic Regression and KNN obtained the same test accuracy, while SVM achieved 66.7%.

5. Business Implication and Limitations
The model can support preliminary launch risk and recovery-cost estimation. However, the dataset is small and imbalanced, so future work should add newer launches, more operational variables, and probability calibration.

Github repository

Deliverable	Repository file
API data collection	jupyter-labs-spacex-data-collection-api.ipynb
Web scraping	jupyter-labs-webscraping.ipynb
Data wrangling	labs-jupyter-spacex-Data-wrangling.ipynb
EDA and feature engineering	eda_visualization.ipynb
SQL analysis	jupyter_labs_eda_sql_coursera_sqllite.ipynb
Folium analysis	lab_jupyter_launch_site_location.ipynb
Plotly Dash	spacex_dash_app.py
Machine learning	SpaceX_Machine_Learning_Prediction.ipynb
Supporting data	dataset_part_1/2/3.csv; Spacex.csv; dash/geo/web CSV files
Final presentation	SpaceX_Capstone_Final_Presentation.pdf

Repository URL: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone